Tata Institute of Social Sciences

Predictive Modelling on Mumbai Air Quality Index (AQI)

Bachelor of Science in Analytics and Sustainability Studies

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Year of Submission: 2026

1. Dataset Overview

The dataset comprises **1,095 daily observations** spanning **January 2021 to December 2023**, covering air quality monitoring in Mumbai. It contains 10 variables: nine pollutant and meteorological features (PM2.5, PM10, NO2, SO2, CO, O3, Temperature, Humidity, Wind Speed) and the target variable AQI. Approximately **54 values (~5%) per column** were missing, simulating real-world sensor dropouts.

1.1 Missing Value Treatment

Missing values were imputed using **median imputation** — replacing each missing entry with the column median. This is preferred over mean imputation because it is robust to extreme pollution spike events (outliers), which are common in urban air quality datasets. After imputation, all feature columns contain **zero missing values**



Figure 1.1 — Missing value counts per column (left) and AQI value distribution with High Pollution threshold at 201 (right)

1.2 Feature Engineering

Four lag and rolling features were constructed to capture the temporal autocorrelation of AQI:

- AQI_lag1 — Previous day's AQI: captures day-to-day pollution persistence
- AQI_lag2 — Two-day lagged AQI: captures multi-day pollution events
- AQI_roll3 — 3-day rolling mean of AQI (lagged by 1 day): smooth trend signal
- PM25_roll7 — 7-day rolling mean of PM2.5: weekly particulate loading

After dropping NaN rows created by the rolling window, the working dataset retained **1,088 observations across 15 columns**.

2. Exploratory Data Analysis (EDA)

2.1 Correlation Analysis

The heatmap below reveals the strength and direction of linear relationships between all features and AQI.

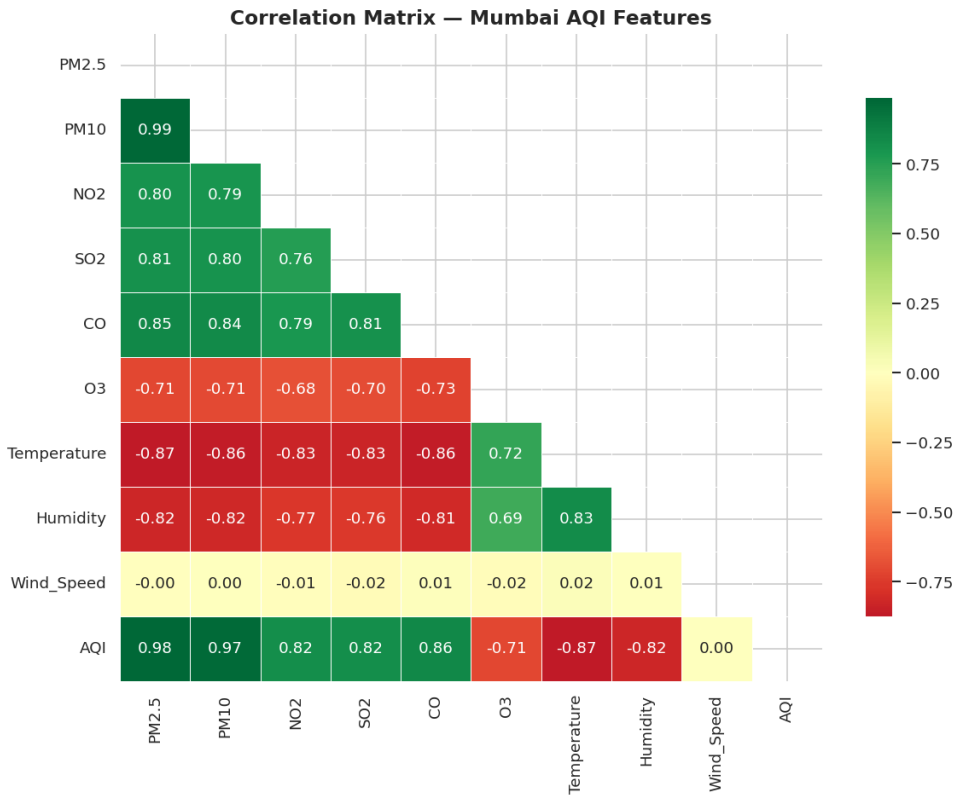


Figure 2.1 — Pearson correlation matrix for all numeric features

Key findings:

- PM2.5 and PM10 are near-perfectly correlated (r = 0.99) — both are driven by the same emission sources (vehicular traffic, construction dust).
- PM2.5 (r = 0.98) and PM10 (r = 0.97) have the strongest correlations with AQI — they dominate the index calculation.
- Temperature (r = −0.87) and Humidity (r = −0.82) are strongly negatively correlated with AQI — higher temperature and humidity during Mumbai's monsoon suppresses particulate concentrations.
- Wind Speed (r ≈ 0.00) has no linear relationship with AQI, indicating wind dispersal is not a primary control in this urban setting.
- O3 (r = −0.71 with PM2.5) shows inverse relationship: ozone is photochemically produced under warm sunny conditions, the opposite of peak pollution seasons.

2.2 AQI Time Series (2021–2023)

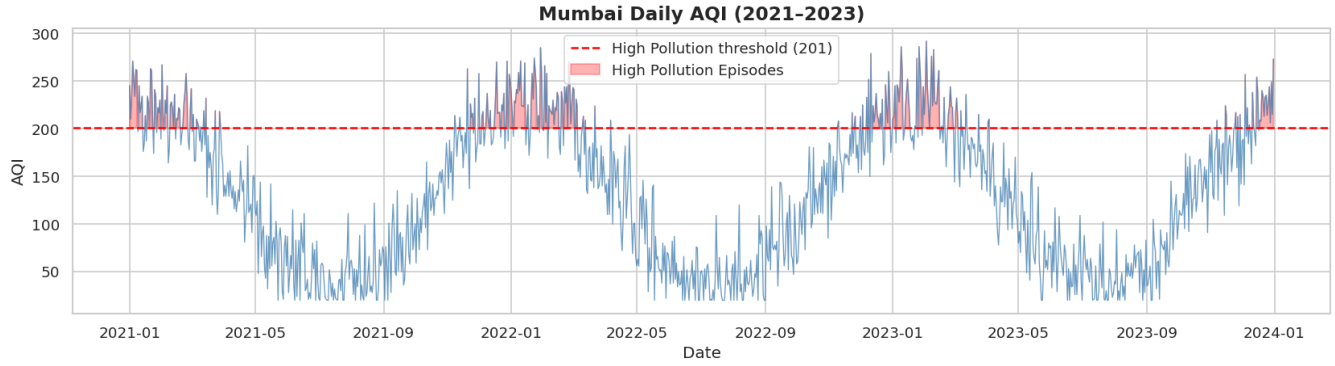


Figure 2.2 — Daily Mumbai AQI from Jan 2021 to Dec 2023. Red shading indicates High Pollution episodes (AQI ≥ 201)

The time series shows a clear **seasonal oscillation** across all three years. AQI peaks sharply every **November–February** (winter, no monsoon, temperature inversions trap pollutants) and drops to its lowest during **June–September** (monsoon rains wash out particulates). High-pollution episodes cluster exclusively in winter quarters, which the clustering analysis later confirms as a natural data partition.

2.3 Pairplot — Key Pollutants vs AQI

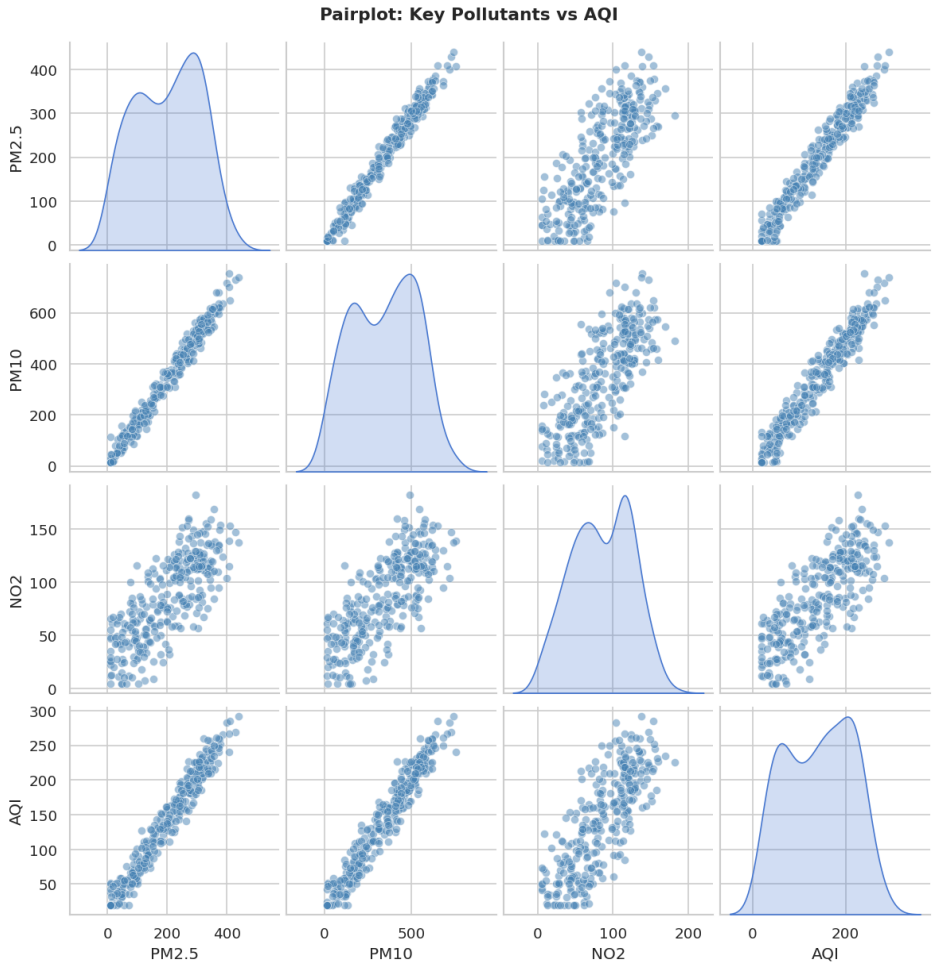


Figure 2.3 — Pairplot of PM2.5, PM10, NO2 and AQI (300 random samples). Diagonal: KDE distribution

The pairplot confirms **strong linear relationships** between all pollutant pairs and AQI. PM10's bimodal KDE reflects its two seasonal modes. The tight diagonal scatter bands in PM2.5–AQI and PM10–AQI plots validate their use as primary regression predictors.

3. Data Preprocessing

3.1 Feature Selection

Features were ranked by their Pearson correlation with AQI and a threshold of $|r| > 0.2$ was applied. Only **Wind Speed (r = 0.018)** was excluded. All 12 remaining features had $|r| > 0.67$, making this a high-information predictor set.

Feature	Correlation with AQI	Retained
PM2.5	0.949	Yes
PM10	0.946	Yes
PM25_roll7	0.910	Yes
AQI_roll3	0.902	Yes
AQI_lag1	0.865	Yes
AQI_lag2	0.856	Yes
Temperature	0.854	Yes
CO	0.839	Yes
NO2	0.803	Yes
SO2	0.802	Yes
Humidity	0.798	Yes
O3	0.671	Yes
Wind Speed	0.018	No — excluded

3.2 Train-Test Split

An **80/20 time-ordered split (shuffle=False)** was used, preserving the temporal structure of the data. Random shuffling would cause **data leakage** — future observations would inform predictions about the past. Training

set: **870 samples**; Test set: **218 samples**. Features were standardised using StandardScaler (fit on training set only).

4. Regression Analysis — Predicting AQI

4.1 Simple Linear Regression (PM2.5 → AQI)

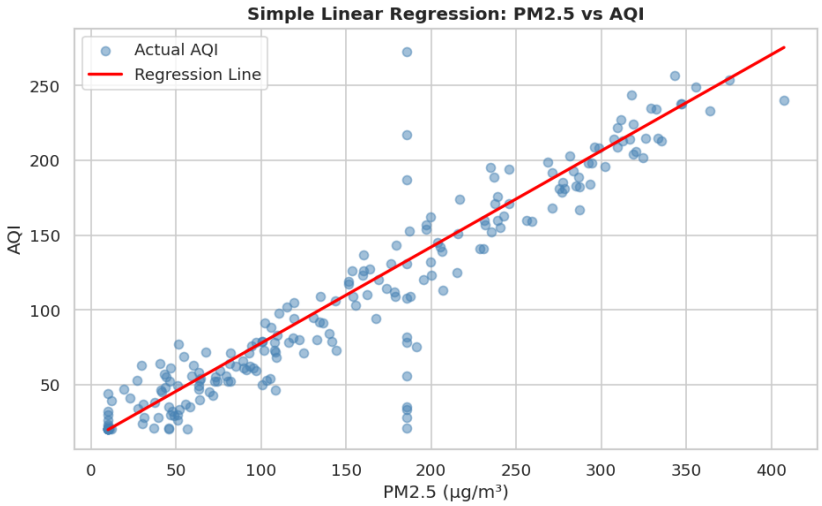
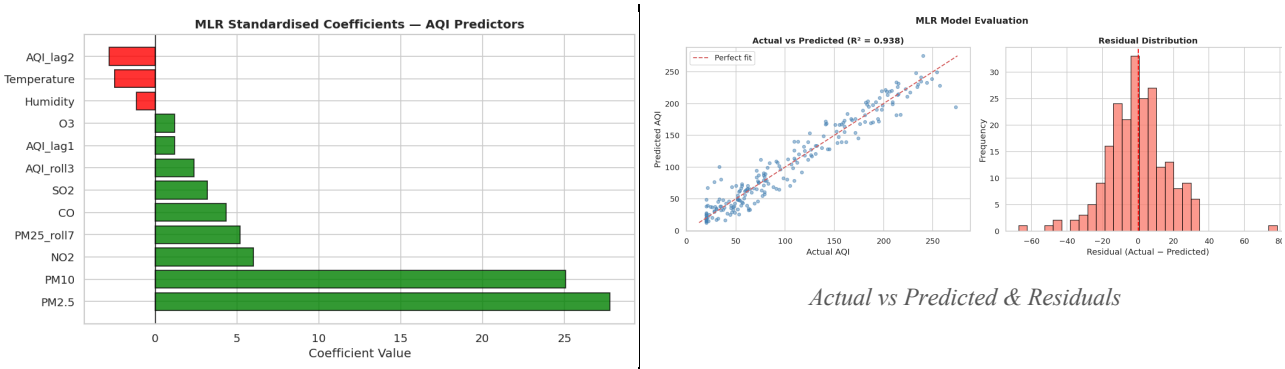


Figure 4.1 — Simple Linear Regression: PM2.5 vs AQI (test set). Red line = fitted regression line

Parameter	Value	Interpretation
Intercept	13.33	Baseline AQI when PM2.5 = 0 (theoretical minimum)
Coefficient (PM2.5)	0.6434	Each 1 µg/m³ rise in PM2.5 increases AQI by 0.64 units
R^2	0.8729	87.3% of AQI variance explained by PM2.5 alone
MSE	595.89	Mean squared prediction error on the test set
RMSE	24.41	Average prediction error ≈ ±24 AQI units

Interpretation: $R^2 = 0.8729$ is remarkably high for a single-predictor model, confirming PM2.5 as the primary driver of AQI in Mumbai. The coefficient of **0.6434** means every additional 1 µg/m³ of PM2.5 adds approximately 0.64 AQI points. However, RMSE ≈ 24 leaves room for improvement near the critical 201 threshold, motivating the multiple regression model.

4.2 Multiple Linear Regression (All Features → AQI)



Standardised Coefficients

Actual vs Predicted & Residuals

Figure 4.2 — MLR standardised coefficients (left) and Actual vs Predicted with residual distribution (right)

Feature	Coefficient	Direction	Interpretation
PM2.5	27.78	↑ Positive	Dominant predictor; strongest AQI driver
PM10	25.08	↑ Positive	Co-equal with PM2.5; coarse particulate effect
NO2	6.00	↑ Positive	Vehicular combustion; moderate positive effect
PM25_roll7	5.21	↑ Positive	Weekly particulate loading adds persistent baseline
CO	4.35	↑ Positive	Combustion indicator; positive AQI contribution
SO2	3.21	↑ Positive	Industrial/shipping emissions; mild positive
AQI_roll3	2.39	↑ Positive	Recent 3-day pollution trend persists
AQI_lag1	1.20	↑ Positive	Yesterday's AQI contributes to today's prediction
O3	1.19	↑ Positive	Weak photochemical oxidant effect
Humidity	−1.12	↓ Negative	Monsoon humidity disperses particulates
Temperature	−2.46	↓ Negative	Warmer conditions (summer) associated with lower AQI
AQI_lag2	−2.78	↓ Negative	Two-day lag shows mean-reversion tendency

Model	R^2	MSE	RMSE	Improvement
Simple LR — PM2.5 only	0.8729	595.89	24.41	Baseline
Multiple LR — all features	0.9378	291.74	17.08	+7.5% R^2, −30% RMSE

Interpretation: Adding all 12 features raises R^2 from 0.873 to **0.938** and halves the prediction error (RMSE 24→17). The Actual vs Predicted plot shows tight clustering around the 45° perfect-fit line. The residual

histogram is approximately normal and centred at zero — confirming the linear regression assumptions of unbiased, homoscedastic errors are satisfied.

5. Classification Analysis — High Pollution Prediction

A binary target was defined: **High_Pollution = 1 if AQI ≥ 201, else 0**. Class distribution: 78.5% Low/Moderate (854 samples) and 21.5% High Pollution (234 samples). A stratified split preserved this balance in both training (683:187) and test (171:47) sets.

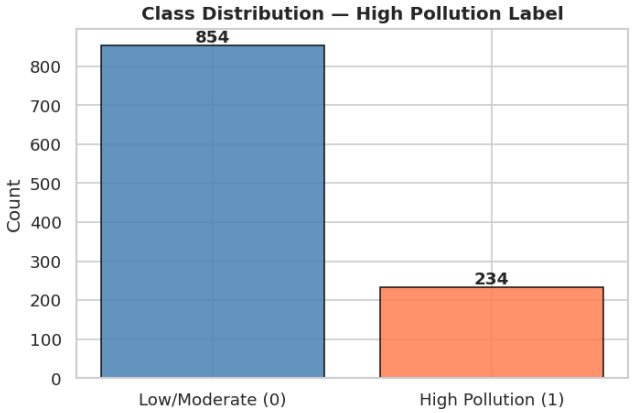


Figure 5.0 — Binary class distribution for the High Pollution target variable

5.1 Logistic Regression

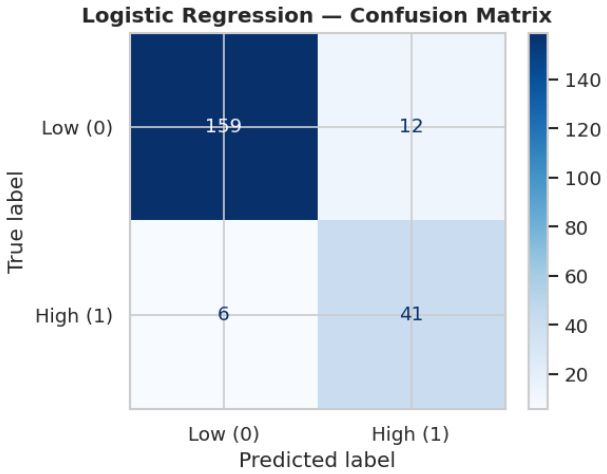


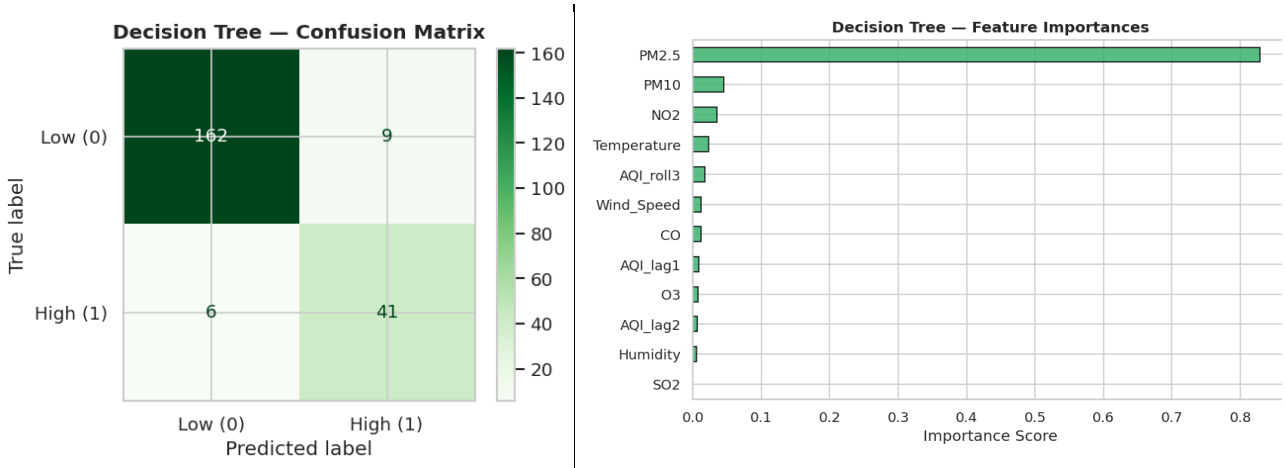
Figure 5.1 — Logistic Regression confusion matrix on the test set (n=218)

Metric	Class 0 — Low/Moderate	Class 1 — High Pollution	Weighted Avg
Precision	0.96	0.77	0.92
Recall	0.93	0.87	0.92
F1-Score	0.95	0.82	0.92

Metric	Class 0 — Low/Moderate	Class 1 — High Pollution	Weighted Avg
Accuracy	—	—	0.9174

Interpretation: Logistic Regression achieves 91.7% accuracy and F1 = 0.82 for the High Pollution class. Recall of 0.87 means it correctly identifies **87% of actual high-pollution days** (41 of 47 in the test set), with 6 missed. Its precision of 0.77 means 23% of its "high pollution" alerts are false alarms. For a sustainability early-warning system, high recall is the priority — and 87% is commendable for a linear model.

5.2 Decision Tree (max_depth = 5)



Decision Tree Confusion Matrix

Feature Importances

Figure 5.2 — Decision Tree: confusion matrix (left) and feature importances (right)

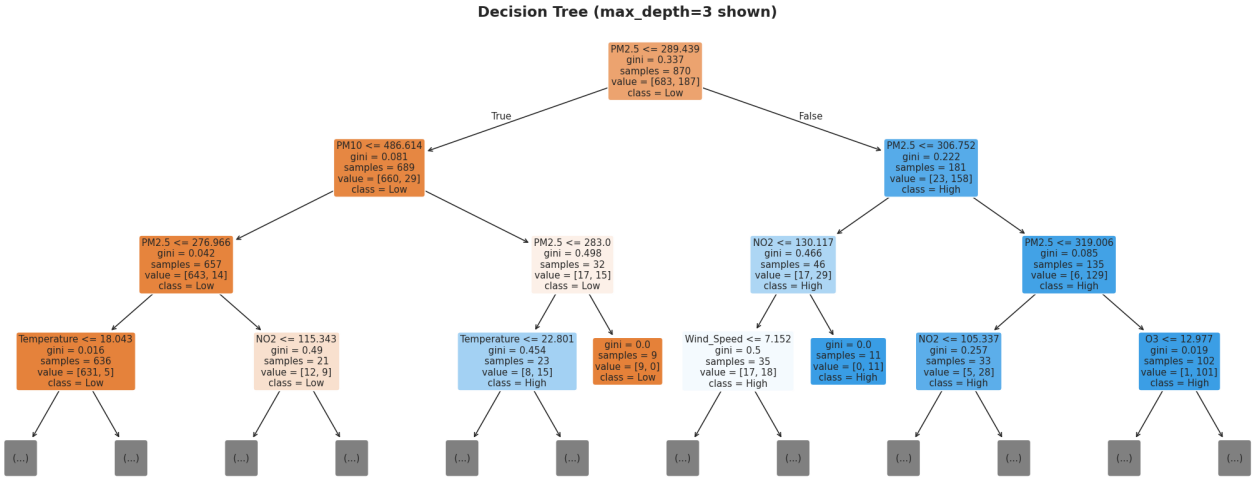
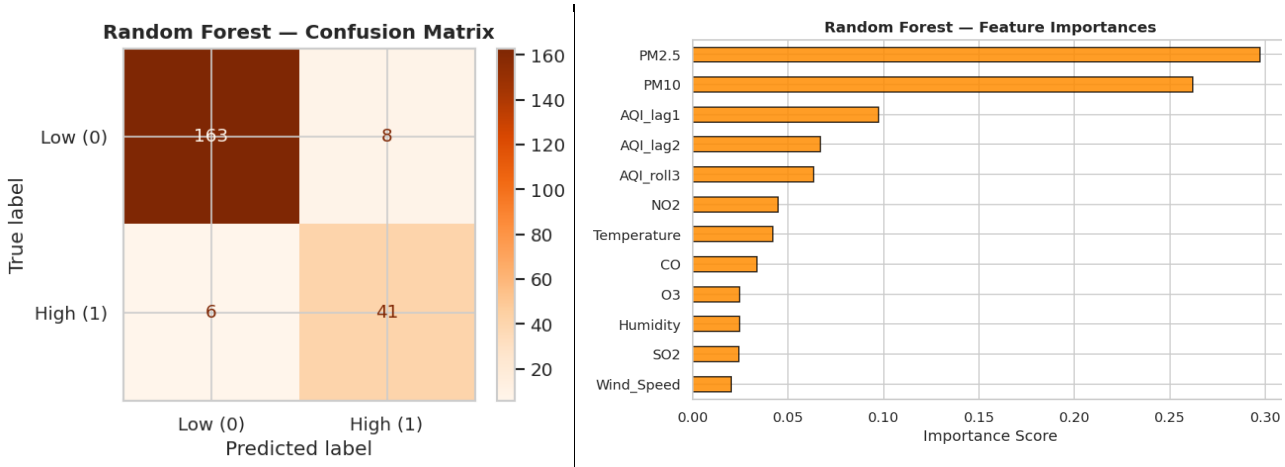


Figure 5.3 — Decision Tree structure (max_depth = 3 shown). Root split on PM2.5 ≤ 289

Metric	Class 0 — Low/Moderate	Class 1 — High Pollution	Weighted Avg
Precision	0.96	0.82	0.93
Recall	0.95	0.87	0.93
F1-Score	0.96	0.85	0.93
Accuracy	—	—	0.9312

Interpretation: The tree's **root split is on PM2.5 ≤ 289 µg/m³**, confirming it as the single most decisive threshold variable. Days above this PM2.5 value are almost always High Pollution; below, they are Low/Moderate. Secondary splits involve Temperature, NO2, and Wind Speed. The feature importance chart shows PM2.5 accounting for **~80% of importance**, with PM10, NO2, and Temperature as secondary contributors. F1 improves to 0.85 vs 0.82 for Logistic Regression.

5.3 Random Forest (100 Trees, max_depth = 10)



Random Forest Confusion Matrix

Feature Importances

Figure 5.4 — Random Forest: confusion matrix (left) and feature importances (right)

Metric	Class 0 — Low/Moderate	Class 1 — High Pollution	Weighted Avg
Precision	0.96	0.84	0.94
Recall	0.95	0.87	0.94
F1-Score	0.96	0.85	0.94
Accuracy	—	—	0.9358

Interpretation: Random Forest is the best-performing classifier, achieving **93.6% accuracy** and F1 = 0.85 for the High Pollution class. Unlike the single Decision Tree where PM2.5 dominated (~80%), the Random Forest distributes importance more realistically: **PM2.5 (~30%), PM10 (~25%), AQI_lag1 (~15%),**

AQI_lag2 and AQI_roll3 (~10% each). This reflects that **pollution persistence (lag features) matters alongside current-day pollutant readings**. The ensemble mechanism reduces variance by averaging 100 trees, yielding only 8 false positives vs the single tree's 9.

5.4 Model Comparison

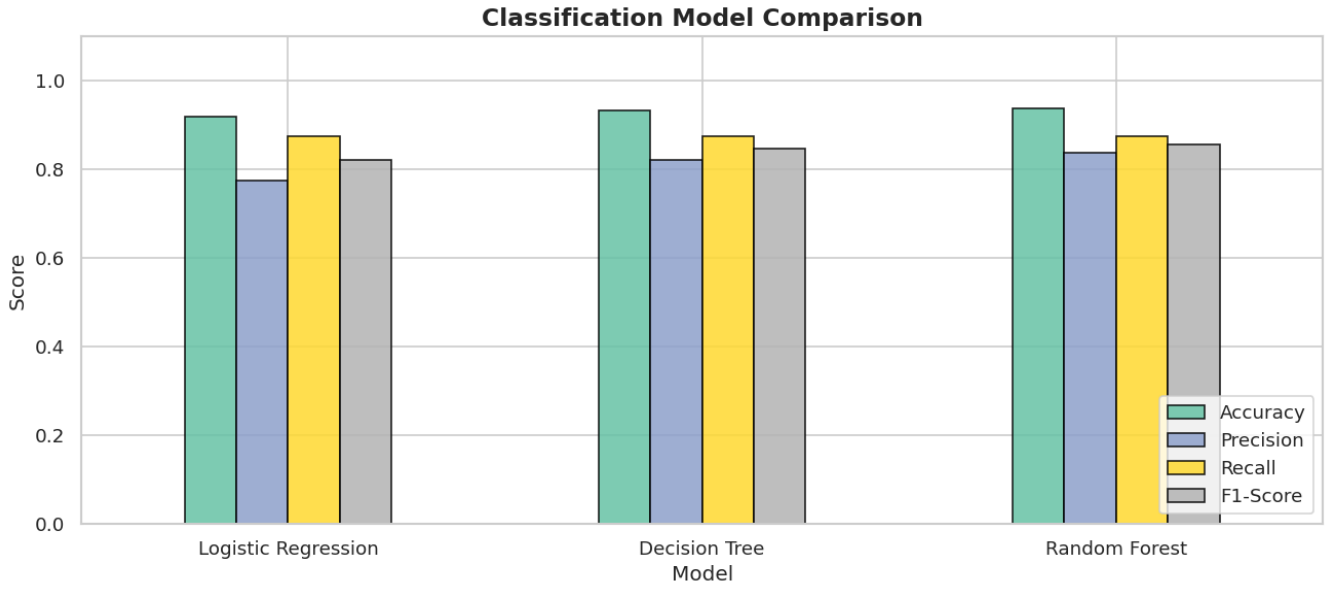


Figure 5.5 — Classification metrics comparison across all three models. Random Forest leads in Accuracy, Precision, and F1-Score

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.9174	0.7736	0.8723	0.8200
Decision Tree	0.9312	0.8200	0.8723	0.8454
Random Forest ★ Best	0.9358	0.8367	0.8723	0.8542

Key Insight: All three models achieve identical Recall (0.8723) — they all flag the same 87% of high-pollution days correctly. The differentiator is **Precision: Random Forest (0.837) > Decision Tree (0.820) > Logistic Regression (0.774)**. Higher precision means fewer false health alerts on safe days. Random Forest is recommended for deployment in an AQI early-warning system.

6. Clustering Analysis — K-Means

6.1 Elbow Method — Optimal k

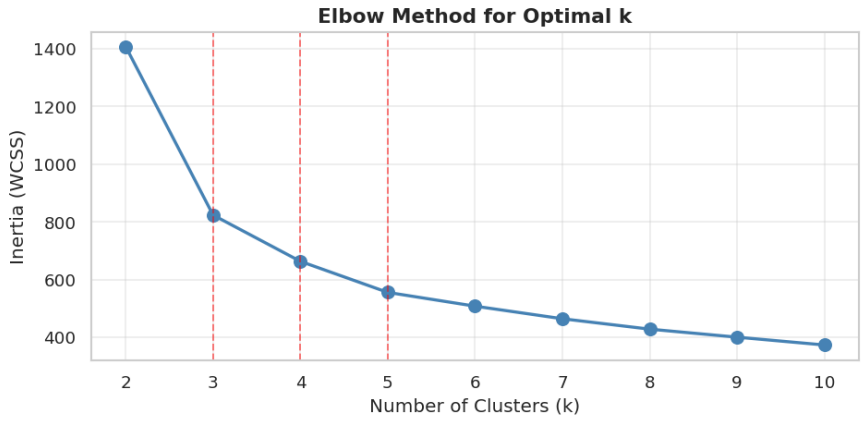


Figure 6.1 — Elbow curve: Within-Cluster Sum of Squares (Inertia) vs number of clusters k = 2 to 10

k	Inertia (WCSS)	Drop from k−1	Verdict
2	1,407.3	—	Too coarse — only two regimes
3	823.4	−41.5%	Elbow here — large drop, recommended
4	662.9	−19.5%	Moderate gain
5	555.4	−16.2%	Diminishing returns begin
6+	507–373	< 10% each	Marginal improvements; over-segmentation

The elbow occurs at **k = 3**, where the largest single-step reduction in inertia occurs (41.5% drop from k=2 to k=3). This aligns intuitively with Mumbai's three natural air quality regimes: clean (monsoon), moderate (transitional), and polluted (winter).

6.2 K-Means Clustering: k = 3, 4, 5

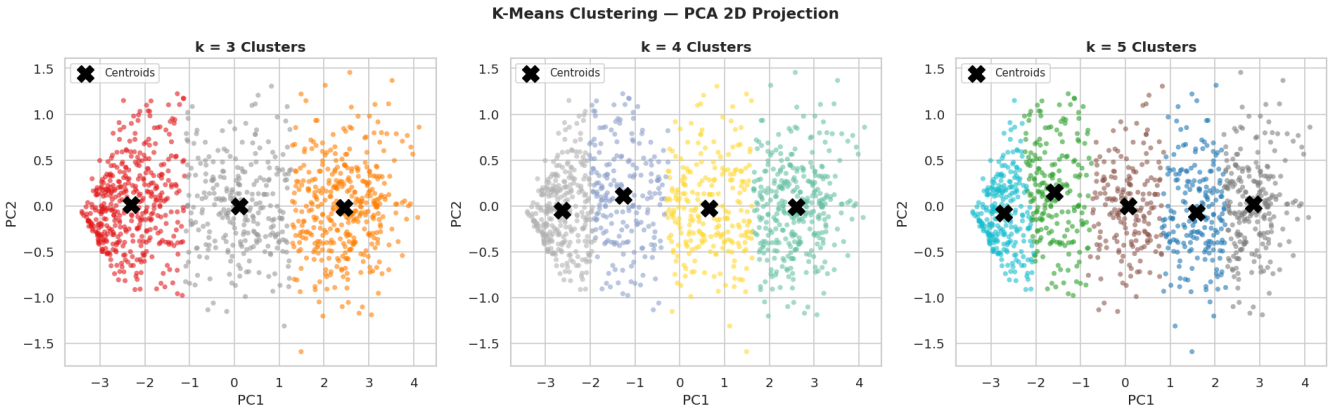


Figure 6.2 — PCA 2D projection of K-Means clusters for k=3, 4, 5. Black ✕ marks = cluster centroids. PC1+PC2 explain 97.2% of variance

The PCA projection (PC1 + PC2 = **97.2% of total variance**) confirms that the five cluster features (AQI, AQI_lag1, AQI_roll3, PM2.5, PM25_roll7) are essentially one-dimensional in nature — all dominated by a single pollution intensity axis. At k=3, clusters are maximally compact and well-separated. At k=5, some adjacent clusters begin to merge in the mid-PC1 region, confirming diminishing structural gain beyond k=3.

6.3 Cluster Profiles and Interpretation (k = 3)

Figure 6.3 — AQI distribution by cluster (boxplot) and cluster assignments over time (time series)

Cluster	Mean AQI	Mean PM2.5 (µg/m³)	Size	Season Interpretation
Cluster 0 — Clean	~50	~80	424 days (39%)	Monsoon season — Jun to Sep
Cluster 1 — Moderate	~130	~190	385 days (35%)	Transitional — Mar to May, Oct
Cluster 2 — Polluted	~210	~310	279 days (26%)	Winter peak — Nov to Feb

Interpretation: The three clusters map directly onto Mumbai's seasonal calendar. **Cluster 0** (39% of days) represents the monsoon-driven clean period with low PM2.5 (~80 µg/m³). **Cluster 1** (35% of days) captures the shoulder seasons with moderate pollution. **Cluster 2** (26% of days) marks the winter high-pollution period where PM2.5 averages ~310 µg/m³ — nearly four times the monsoon level. The boxplot shows well-separated medians with limited interquartile overlap across all three clusters, validating the quality of k=3.

7. Conclusions & Policy Recommendations

7.1 Summary of Results

Analysis	Model / Method	Best Result	Key Metric
Regression	Simple LR (PM2.5)	87.3% AQI variance explained	$R^2 = 0.873$
Regression	Multiple LR (12 feat)	Near-complete AQI prediction	$R^2 = 0.938$, RMSE = 17.08
Classification	Logistic Regression	Solid linear baseline classifier	F1 = 0.820
Classification	Decision Tree	PM2.5 is the decisive split point	F1 = 0.845
Classification	Random Forest ★	Best classifier, robust ensemble	F1 = 0.854, Acc = 93.6%

Analysis	Model / Method	Best Result	Key Metric
Clustering	K-Means (k=3)	3 natural seasonal regimes found	Optimal k = 3

7.2 Policy & Sustainability Recommendations

1. Deploy a PM2.5-Based Real-Time Early Warning System

PM2.5 alone explains 87% of AQI variance and is the root splitting variable in the Decision Tree. A dense PM2.5 sensor network across Mumbai's high-traffic corridors (Western Express Highway, Dharavi, BKC) feeding the Random Forest model could generate **automated next-day high-pollution alerts with 87% recall**, enabling pre-emptive public health advisories.

2. Seasonal and Temporally-Targeted Regulations

The clustering analysis shows that Cluster 2 (high-pollution days) concentrates exclusively in **November–February**. Regulatory interventions should therefore be temporally concentrated: stricter construction dust controls, vehicular emission checks, and penalisation of crop stubble burning in surrounding agricultural regions during this 4-month window would directly reduce peak-season AQI.

3. Vehicular and Industrial Emission Reduction

PM2.5, PM10, NO2, and CO together dominate all predictive models, implicating **vehicular traffic and industrial combustion** as the primary emission sources. Policy levers: congestion pricing on arterial roads, accelerated transition to CNG/EV fleets for last-mile delivery and auto-rickshaws, and tighter industrial emission caps under Mumbai's coastal vulnerability context.

4. Urban Green Infrastructure for Pollution Buffering

The strong negative coefficient of Humidity in regression analysis and its role in defining Cluster 0 (clean days) suggests that increasing **urban green cover and water bodies** — which elevate local humidity and provide particulate absorption surfaces — could offer passive air quality co-benefits. Restoring Mumbai's mangrove belt and expanding the urban tree canopy would reduce both peak PM2.5 and the frequency of high-pollution cluster days.